

Case Study – 2: Campus Mess; Analytics Case Study

Background

You are part of the Mess Management Team at SITM Pune. The campus has 5 colleges sharing one central mess facility:

- **SIDTM** – 450 students
- **SSBF** – 380 students
- **SIBM** – 520 students
- **SIMC** – 320 students
- **SSSS** – 280 students

Your Role: Junior Business Analyst helping the Mess Committee make better decisions

The Problem (Simple & Clear)

The mess is losing money and students are complaining!

Key Issues:

1. **Too much food waste** – throwing away 20% of prepared food daily
2. **High costs** – spending ₹15,000 more than budget every month
3. **Unhappy students** – average satisfaction is only 6/10
4. **Poor planning** – some days too much food, some days too little

Your Mission

Help the mess save money and make students happy using **basic Excel analysis!**

Data Provided

You have **3 months of simple daily data** (March–May 2024) with these columns:

- Date, College, Meal (Breakfast/Lunch/Dinner)
- Students Expected, Students Came, Food Prepared (kg), Food Consumed (kg)
- Cost per kg, Total Cost, Student Rating (1–10)

Business Questions to Solve (Using Basic Excel)

LEVEL 1: Basic Calculations

1. **Calculate daily waste:** Food Prepared – Food Consumed
2. **Calculate waste percentage:** $(\text{Waste} \div \text{Food Prepared}) \times 100$
3. **Find average satisfaction:** Use AVERAGE function
4. **Calculate total monthly costs:** Use SUM function

LEVEL 2: Simple Analysis

5. **Which college wastes most food?** Use college-wise totals
6. **Which meal has highest waste?** Compare Breakfast vs Lunch vs Dinner
7. **What's the cost of waste per day?** $\text{Waste (kg)} \times \text{Cost per kg}$
8. **Find best and worst days:** Use MAX/MIN functions

LEVEL 3: Business Insights

9. **If we reduce waste by 50%, how much money will we save?**
10. **Which college should we focus on first for waste reduction?**
11. **What's the relationship between attendance and waste?**
12. **How can we improve student satisfaction?**

Excel Functions You'll Use (Basic Only!)

- **SUM()** – Add up numbers
- **AVERAGE()** – Find average
- **MAX()/MIN()** – Find highest/lowest
- **IF()** – Simple conditions
- **Basic arithmetic** – +, –, ×, ÷, %
- **Simple charts** – Bar charts, pie charts

Expected Outputs

1. Waste Analysis Report (1 page)

- Total waste per college
- Daily waste trends
- Cost impact of waste

2. Simple Dashboard (1 Excel sheet)

- Key numbers in big boxes
- 2–3 simple charts
- Traffic light system (Red/Yellow/Green)

3. Business Recommendations (5 simple points)

- What to do immediately
- Which college to target first
- How much money can be saved

Success Metrics

- Can you identify the biggest waste problem?
- Can you calculate potential savings?
- Can you make simple but useful charts?
- Can you give practical recommendations?